Data Visualisation Final Report
WHO Retrospective: The MDG Era 2000–2015

By Nomaer I. R. Naiem

1. Introduction

This project presents a data visualisation solution built on the WHO Life Expectancy dataset compiled by Rajarshi (2018) on Kaggle, which combines health, demographic and economic indicators for 193 countries from 2000 to 2015. The dataset was chosen because it covers the precise time window of the United Nations Millennium Development Goals (MDGs), making it uniquely suited to a retrospective analysis of where health gains were greatest, which countries lagged, and which interventions most plausibly explain the differences.

1.1 Data Dictionary

The dataset contains 22 attributes; the subset below was used in this project. The key variables used in the dataset are listed in the table below. There are some additional Derived variables which were used and are also included.

Variable	Type	Description	Derived?
Country	Dimension (Nominal)	Country names	No
Year	Dimension (Date Converted)	Year of observation	No
Status	Dimension (Binary)	Developed, Developing	No
Life expectancy	Measure (Continuous)	Life expectancy in years	No
HIV/AIDS	Measure (Continuous)	Deaths per 1,000 due to HIV/AIDS	No
Schooling	Measure (Continuous)	Average number of years of schooling	No
Total Expenditure	Measure (Continuous)	Government health expenditure as % of total expenditure	No
Avg Expenditure in 15 Years	Measure (Continuous)	Each country's mean of Total Expenditure across 2000–2015	Yes
LE Change in 15 Years	Measure (Continuous)	Each country's Life expectancy in 2015 minus its value in 2000	Yes
Schooling Change in 15 Years	Measure (Continuous)	Each country's Schooling in 2015 minus its value in 2000	Yes

Cohort	String	Categorises each country as "Lagger countries (LE < 60 in 2000)", "Other developing", or "Developed" using a FIXED LOD expression	Yes

Two derived measures were computed for the complex question: LE Change in 15 Years and Schooling Change in 15 Years (each country's value in 2015 minus 2000), together with Avg Expenditure in 15 Years (the country mean of Total expenditure). These are computed in Tableau using LOD expressions; the Plotly Dash implementation derives them in pandas before plotting.

1.2 Persona and Questions

Persona: “Dr Amara Okonkwo, Senior Policy Analyst, WHO Department of Health Systems and Innovation. She is preparing a post-MDG retrospective briefing for the 69th World Health Assembly in May 2016. Her audience comprises senior delegates and ministers of health with strong domain knowledge but no statistical training. She needs a single, narratively coherent dashboard that she can present from and that delegates can explore. She works on a 1920×1080 conference room display.”

To answer the research question, three simple and one complex question have been formulated as follows.

- SQ1: How did average life expectancy evolve in developed versus developing countries between 2000 and 2015, and did the gap narrow?
- SQ2: How is life expectancy distributed across the world, and which regions or countries consistently lag?
- SQ3: For the lagging countries identified in SQ2, how did HIV/AIDS-related deaths change between 2000 and 2015, and did this differ from the rest of the developing world?
- CQ: Among the lagging countries, what combination of HIV/AIDS reduction, schooling investment and health expenditure separated the strongest gainers in life expectancy from those that stagnated or declined?

1.3 Requirements

Translating each question into a functional requirement for the Dashboard:

- R1 (relates to SQ1): The user needs to compare the trajectories of developed and developing countries on a common time axis to assess whether the global gap narrowed during the MDG era.
- R2 (relates to SQ2): The user needs to identify the geographic distribution of life expectancy and pinpoint the regions that consistently lag, thereby providing the spatial baseline that anchors SQ3 and the CQ.
- R3 (relates to SQ3): The user needs to compare the HIV/AIDS death trajectory of the lagging cohort with those of the rest of the developing world and the developed world, so that the

magnitude of the lagging reduction can be assessed against an explicit baseline rather than asserted in isolation.

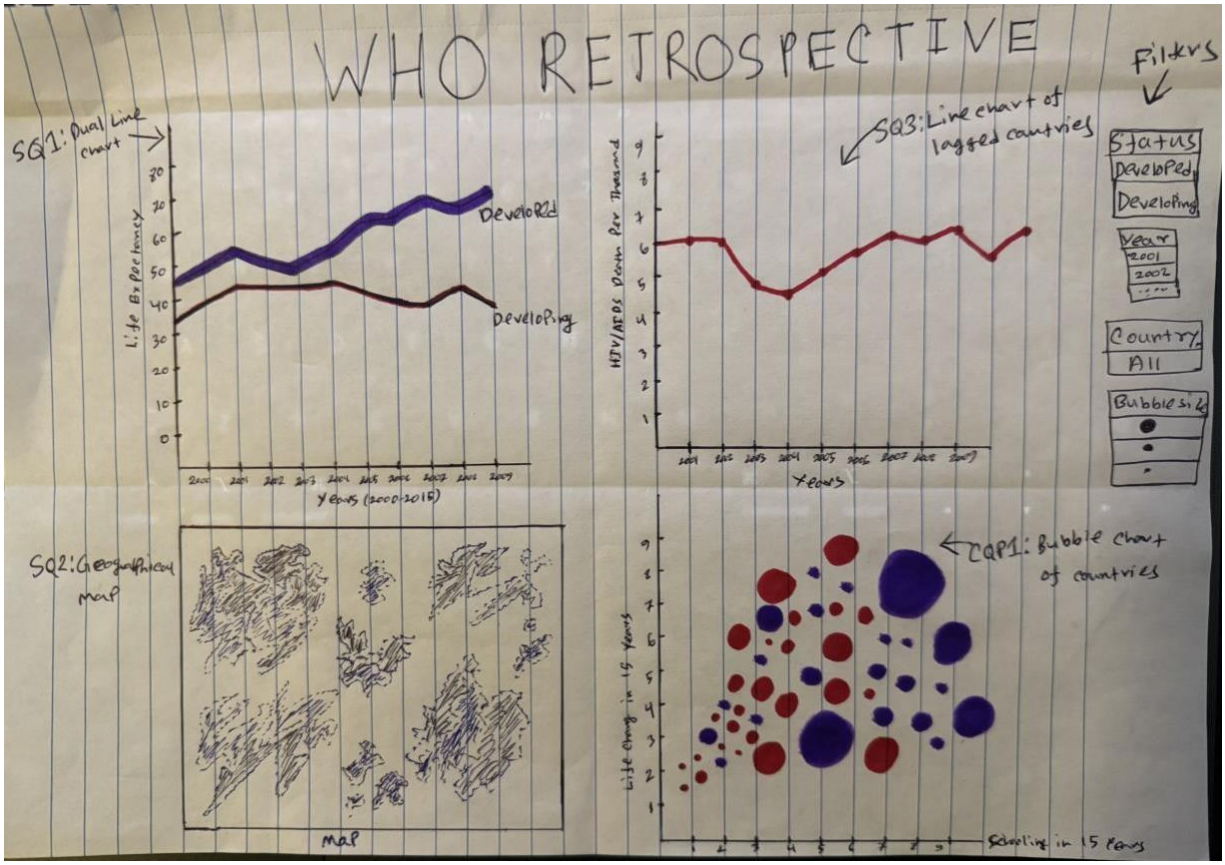
- R4 (relates to CQ): The user needs to explore the multivariate relationship among HIV reduction, schooling investment and health expenditure for the lagging cohort, and identify which countries combined these levers most successfully.
- R5 (Coordination and presentation): All four views must share the Country dimension as a coordinated filter, so that selecting any country in either the SQ2 map or the CQ P1 bubble chart filters the other three views to that country. The dashboard layout must respect the 1920×1080 conference room display specified by the persona, and chart titles must be phrased as the underlying questions so the dashboard remains readable without external instructions (Nielsen, 1994).

2. Design

2.1 Paper Landscape

Following the user-centred design practice recommended by Few (2009) and the Tufte (2001) principle that design should precede tooling, the dashboard was first sketched on paper. The sketch establishes the four chart types, the spatial hierarchy, the legend column on the right, and the role of each filter.

Figure 1: Paper Landscape

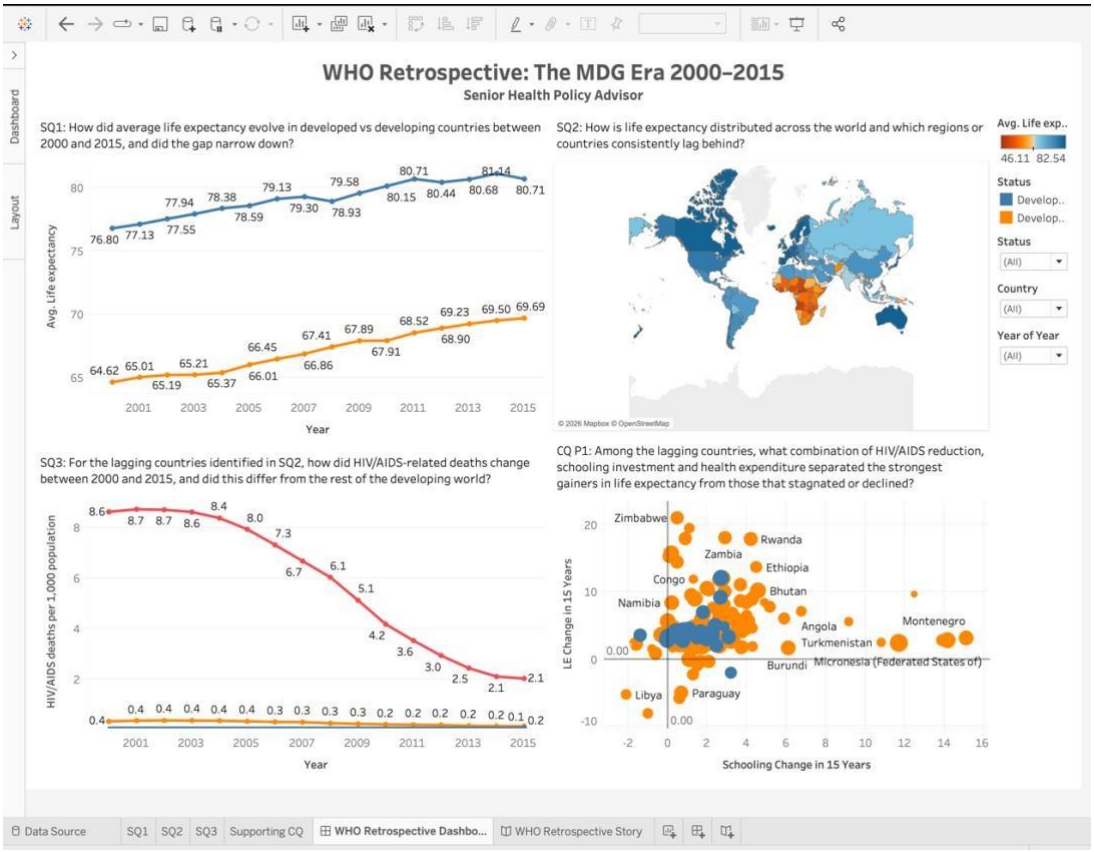


The paper landscape follows. SQ1 (top-left, dual-line) anchors the global comparison; SQ2 (bottomleft, choropleth) localises it; SQ3 (top-right) introduces the HIV story; and CQP1 (bottom-right) supports the complex question.

2.2 Final Dashboard

Interaction guide: Click any country on the map (SQ2) or any bubble on CQ P1: all four views will filter to that country. Compare a gainer (e.g., Zimbabwe) with a decliner (e.g., Libya) to see the full pattern across schooling, life expectancy, HIV deaths, and health expenditure.

Figure 2: Final Tableau Dashboard



Screenshots are captured at A4 landscape because the native 1920×1080 dashboard does not fit a single laptop screen; the submitted ". twbx" is set to 1920×1080 as required by the brief.

2.3 Design Evolution from Paper Landscape to Tableau

The differences between Figure 1 and Figure 2 are summarised below; each change reflects an HCI or visualisation principle rather than aesthetic preference.

- Sheet placement swapped (SQ2 moved up, SQ3 down). The final layout follows the natural left-to-right, top-to-bottom reading order described by Wickens (2008): global comparison → geographic distribution → cohort focus → country interaction.
- Quadrant guide-lines added to CQ P1. Dashed reference lines at the global means split the chart into four interpretable quadrants (top-right = gainers; bottom-left = decliners), removing the need for the user to compute means in their head, an application of Few's (2009) principle of removing avoidable cognitive load.

- The SQ1 y-axis did not include 0. Tufte (2001) cautions against truncated axes that exaggerate differences, but life expectancy values never approach zero, and a 0–80 axis would compress the convergence story so heavily that the narrowing gap (12.2 → 11.0 years) would be invisible. Following Few's (2009) more pragmatic position, the axis did not include 0.
- Developed and Other-developing lines were retained on SQ3. Both lines hover near zero, which initially looked like wasted ink (Tufte 2001), but were retained because their flatness is itself the message: the lagging cohort's HIV burden was an order of magnitude higher than that of all other countries, and that gap closed as the laggers improved.
- Bubble-size filter dropped; Country filter retained globally. The sketch's bubble-size filter added complexity without analytical value. Conversely, the Country filter was made global because it is the backbone of the complex-question interaction.
- Direct labelling on SQ1, SQ3 and CQ P1. Value labels were added to the SQ1 lines (showing each year's life expectancy average), each year's cohort average on SQ3, and country labels to the CQ P1 outliers, removing the round-trip from chart to axis and providing the visual anchors the persona needs at a glance (Few, 2009).

3. Implementation

3.1 Tableau

The dataset was loaded into Tableau Public via the CSV connector. Two FIXED-LOD calculated fields underpin all four sheets: Cohort classifies each country as a Lagger (LE < 60 in 2000), Other developing, or Developed; LE Change in 15 Years and Schooling Change in 15 Years compute country-level differences (2015 minus 2000), so each country contributes one value to the bubble chart.

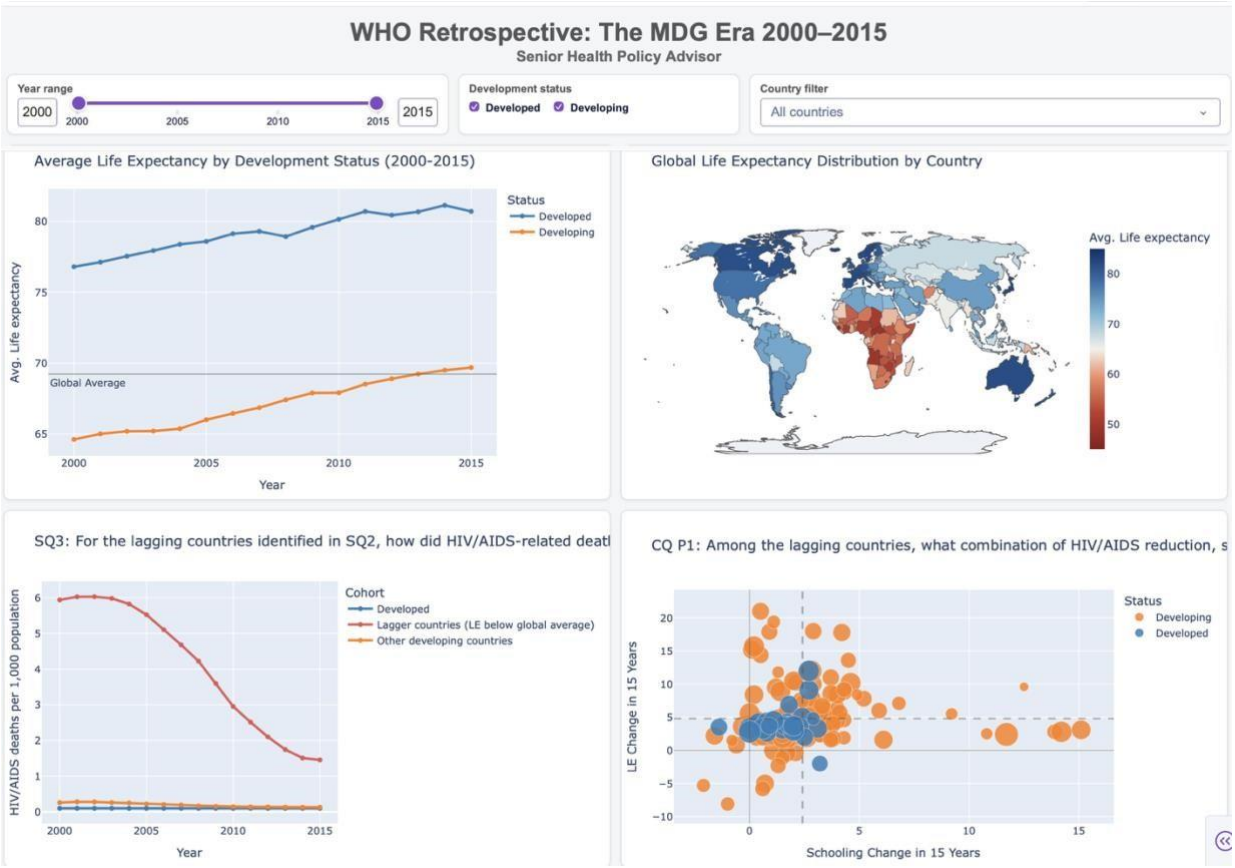
SQ1 places YEAR(Year) on Columns and AVG (Life expectancy) on Rows, with Status on Colour and end-point labelling. SQ2 uses Tableau's geographic role detection and a Red-Blue diverging palette fixed at 46–82; colour value, per Bertin (1983), is the appropriate visual variable for ordered quantitative comparisons. SQ3 places YEAR on Columns and AVG(HIV/AIDS) on Rows, with Cohort on Colour. CQ P1 places Schooling Change on Columns, LE Change on Rows, Avg Expenditure on Size and Status on Colour, with reference lines at the global means defining the four quadrants.

Dashboard composition: Sheets were arranged in a 2×2 grid on a 1920×1080 dashboard. Three filter actions link the views: a Country filter from SQ2 → SQ1, SQ3, and CQ P1; a Country filter from CQ P1 → SQ1, SQ2, and SQ3 (so the user can also spot quadrant outliers); and a Country highlight from CQ P1 → SQ2 that dims unselected countries on the map without filtering them. A Year filter is exposed only on SQ2's sidebar; propagating it globally would collapse the time-series sheets to a single point, and the bubble chart is undefined for a single year.

3.2 Second Implementation: Plotly Dash (Python)

To satisfy the second implementation requirement and triangulate the design choices made in Tableau, a Python implementation was built using Plotly Express for the charts and Dash for the layout, callbacks, and HTTP serving. The same dataset and the same four chart types are used; the differences lie in how the design is expressed and how the user interacts with the result.

Figure 3: Dashboard using Python



The Plotly Dash implementation uses the same four chart types but is driven by an explicit year-range slider, a Status checklist and a Country dropdown rather than by dashboard filter actions.

3.3 Comparison of the Two Implementations

Tableau and Plotly Dash represent two fundamentally different philosophies. Tableau is a directmanipulation environment: the designer drags fields onto shelves, and the system translates that action into a visualisation. Design iteration is therefore cheap; re-colouring a series, swapping the year filter scope, or re-sizing a sheet are one or two clicks. The cost is that customisations beyond the supplied options can be awkward. Plotly Dash inverts the trade-off: every chart, filter, and layout choice exists in code, which makes the design fully scriptable and reproducible but means small changes, such as a legend label, the ordering of two cards, or a colour scale, require editing source files and restarting the server. For an analyst preparing a single, polished briefing, Tableau is the more efficient tool; for a developer embedding a dashboard inside a larger Python application or versioncontrolling its design history, Dash's code-first model is the more suitable one.

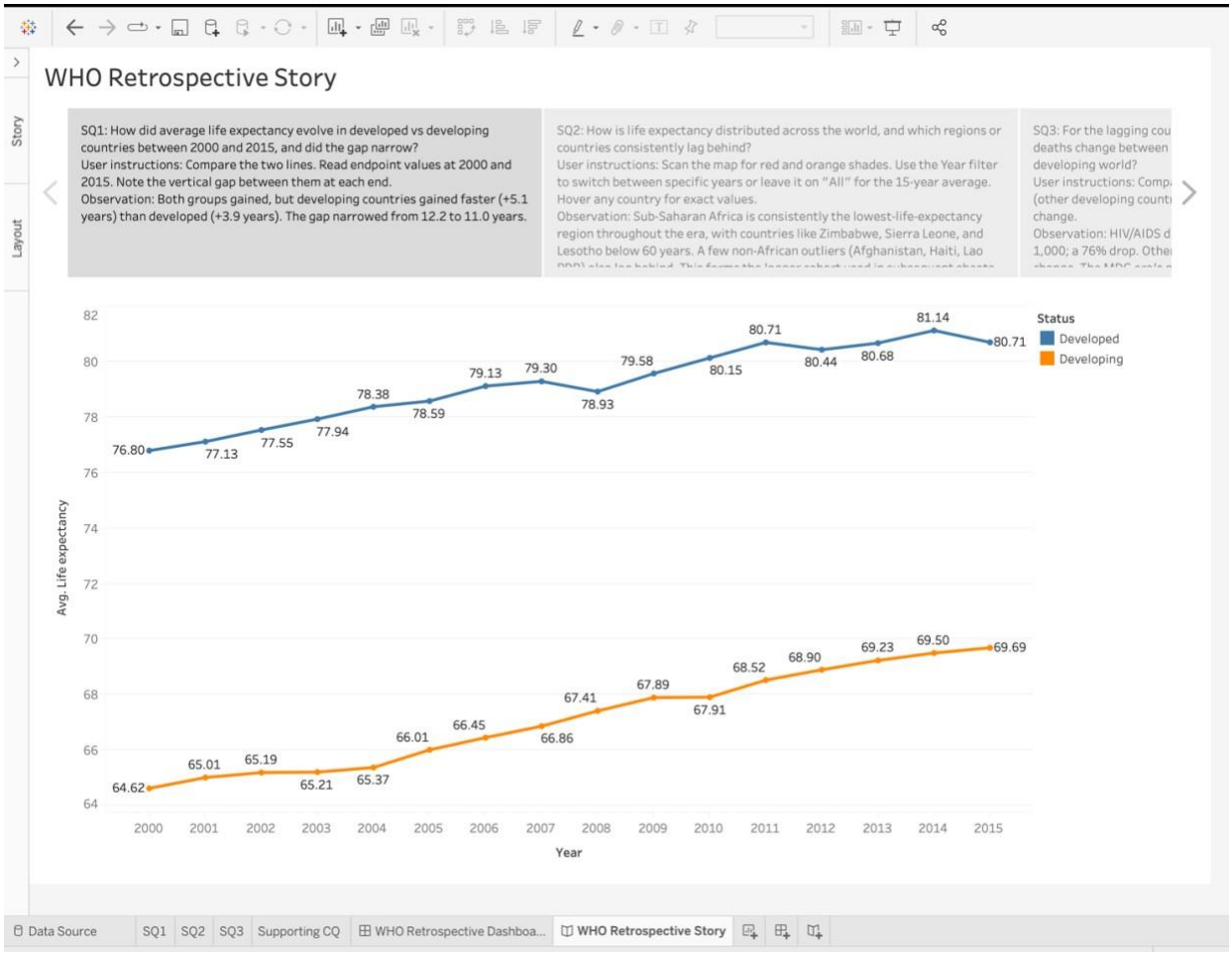
Two specific differences merit comment. First, the Plotly Dash version exposes a year-range slider that lets the user constrain SQ1 and SQ3 to any sub-period within 2000–2015; Tableau's quick-filter equivalent is single-valued for time-series and was therefore omitted. Second, Tableau's geographic role detection automatically resolved country names to map polygons, whereas Plotly Dash required an explicit join to ISO-3 country codes (using pycountry) before px.choropleth would render the map; this illustrates the engineering work made visible in a code-first stack.

4 Walkthrough

The dashboard answers each research question through a corresponding view, and the complex question through interaction across all four. The Tableau Story sheet captures the same five-step narrative for the visualizer.

4.1 SQ1 How did average life expectancy evolve in developed vs developing countries between 2000 and 2015, and did the gap narrow?

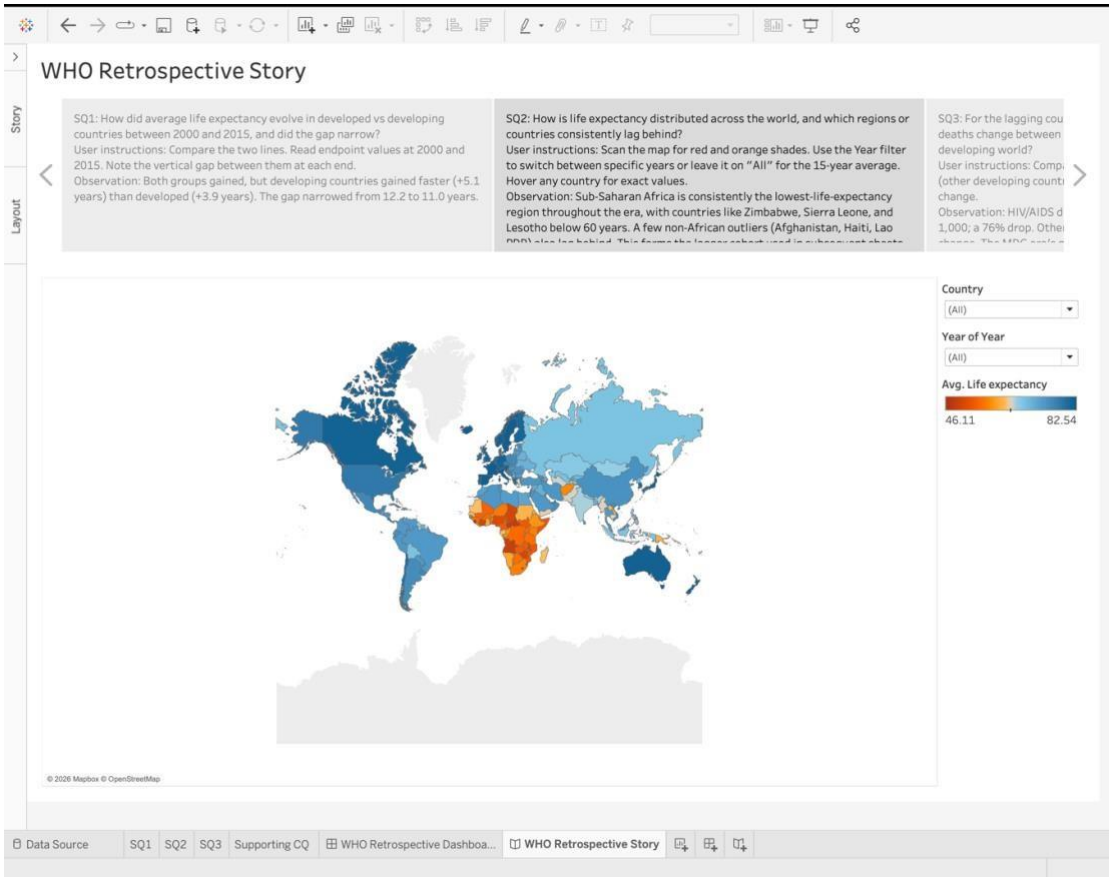
Figure 4: Story Point 1 (SQ1): both groups gain, with developing countries gaining more.



Reading the endpoints in *Figure 4*: developed countries rose from 76.80 to 80.71 years (+3.91 years); developing countries rose from 64.62 to 69.69 years (+5.07 years). The absolute gap, therefore, narrowed from 12.18 to 11.02 years, a real but modest convergence.

4.2 SQ2 How is life expectancy distributed across the world, and which regions or countries consistently lag?

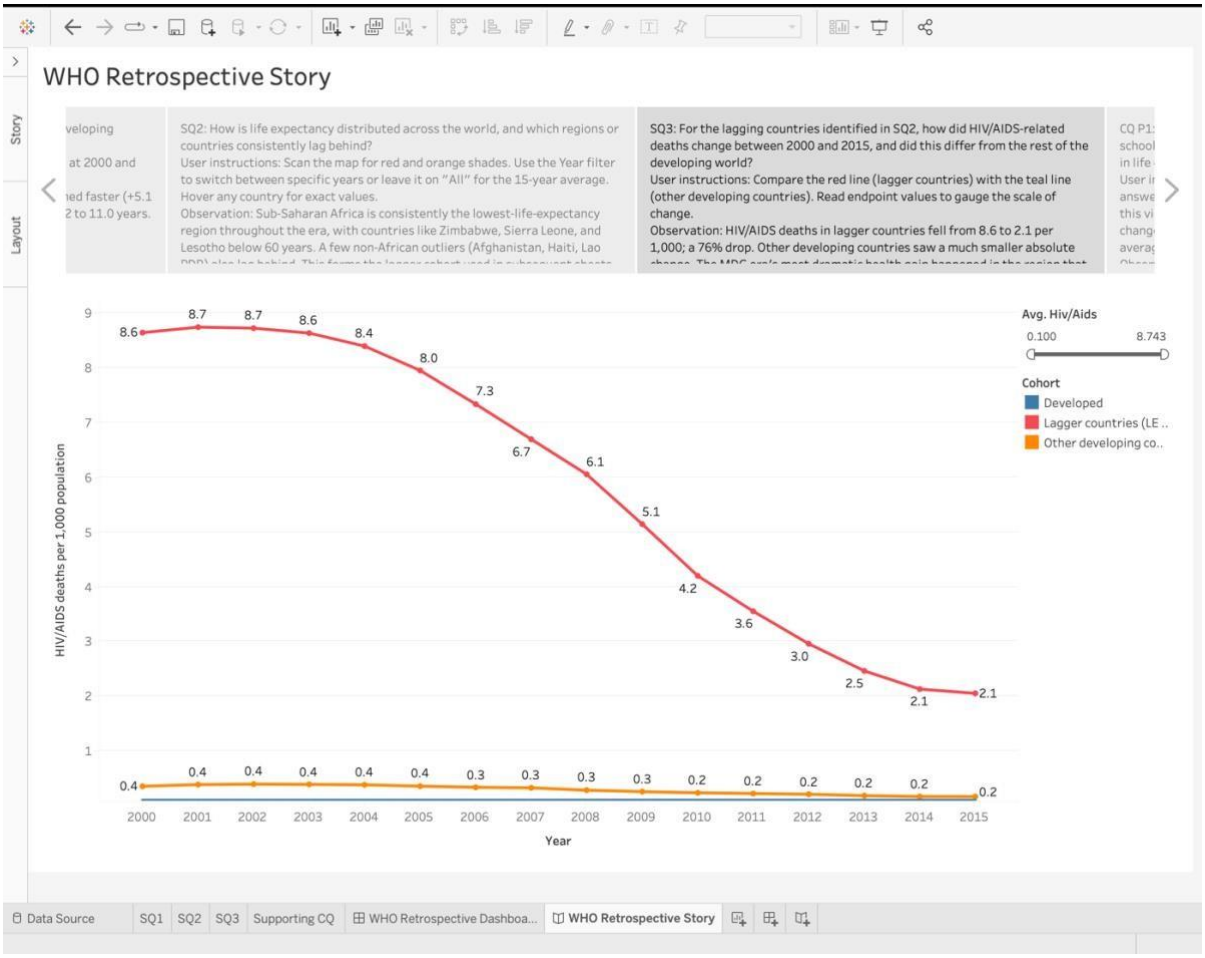
Figure 5: Story Point 2 (SQ2): the geographic distribution of life expectancy.



In *Figure 5*, Sub-Saharan Africa has the lowest life expectancy throughout the period. Sierra Leone, Lesotho, the Central African Republic and Eritrea consistently remain below 60 years; non-African outliers (Afghanistan, Haiti, Lao PDR) fall within the same band. The Year filter allows the user to verify that the geographic pattern is stable across 2000–2015. This forms the lagging cohort referenced in SQ3 and the CQ.

4.3 SQ3 For the lagging countries identified in SQ2, how did HIV/AIDS-related deaths change between 2000 and 2015, and did this differ from the rest of the developing world?

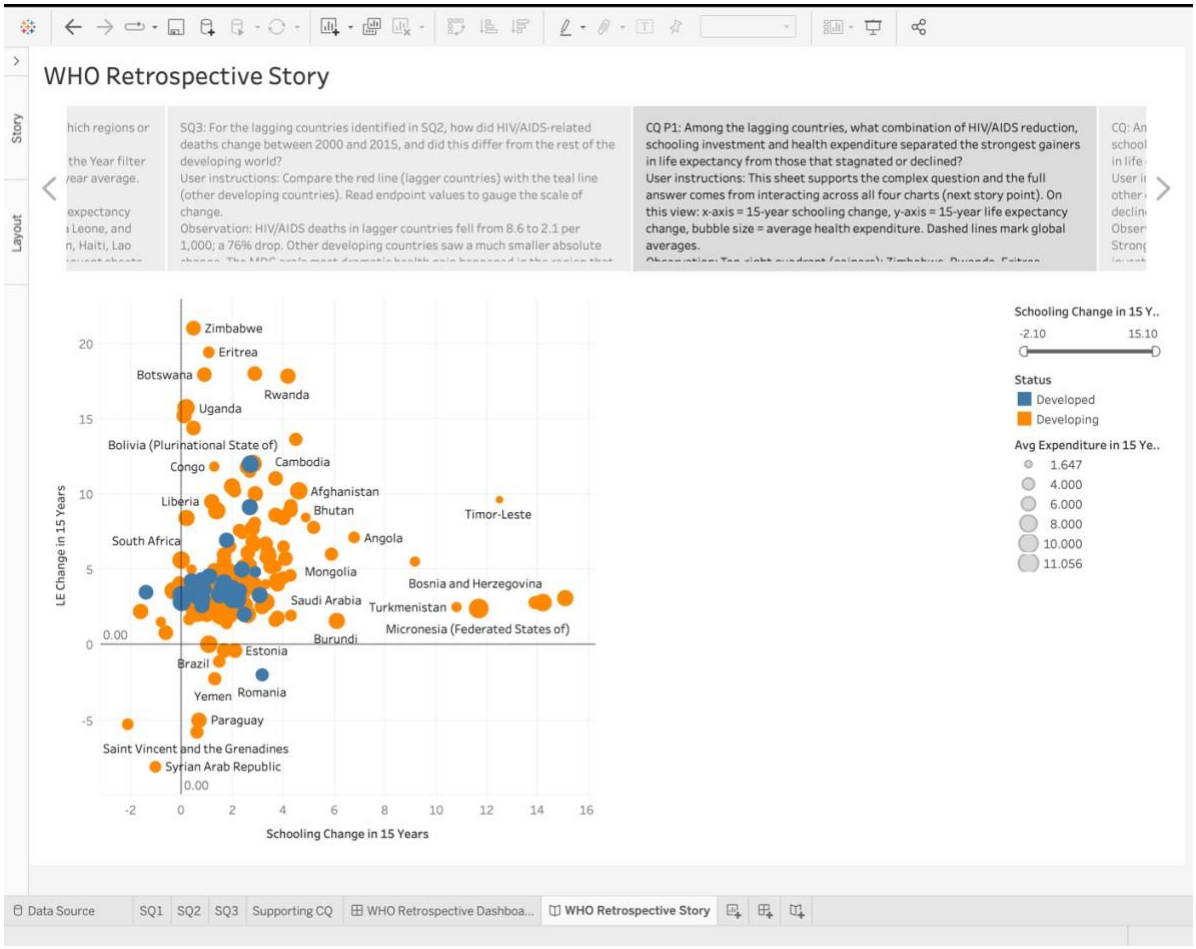
Figure 6: Story Point 3 (SQ3): the lagging HIV reduction (red) against the comparison cohorts.



In *Figure 6*, amongst the 49 lagging countries, average HIV/AIDS deaths fell from 8.6 to 2.1 per 1,000: a 76% reduction. The other developing line is essentially flat at 0.4 → 0.2; the developed line is flat at near-zero. The reduction is therefore specific to the cohort that began the era furthest behind and is plausibly attributable to the international scale-up of antiretroviral therapy under PEPFAR and the Global Fund (Avert, 2024).

4.4 CQ P1 Support for the complex question

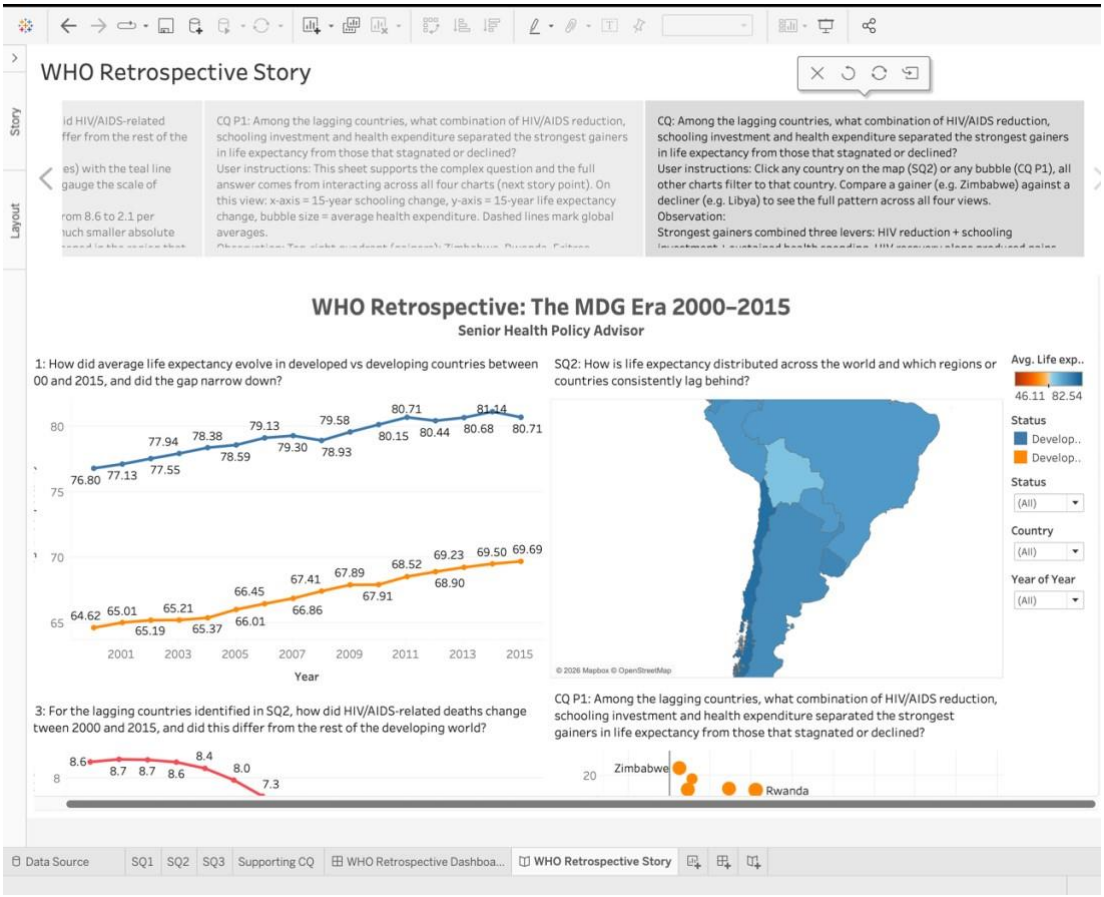
Figure 7: Story Point 4 (CQ P1): the bubble chart used in support of the complex question



In *Figure 7*, the scatter on its own shows that the strongest gains in life expectancy cluster in the righthand half; they also gained in schooling. Zimbabwe, Rwanda, Eritrea, Botswana and Ethiopia occupy the top-right quadrant of gainers; Syria, Libya and Yemen occupy the bottom-left quadrant of decliners. Bubble size shows that gainers tended to maintain comparable or higher health expenditure as a share of GDP.

4.5 Among the lagging countries, what combination of HIV/AIDS reduction, schooling investment and health expenditure separated the strongest gainers in life expectancy from those that stagnated or declined?

Figure 8: Story Point 5 (CQ): the dashboard view, used for country-by-country interaction.



In *Figure 8*, clicking a country in either SQ2 or CQ P1 filters the other three views to that country. Clicking Zimbabwe shows a sharp rise in life expectancy after 2005, steeply falling HIV deaths, and a top-right gainer position. Clicking Libya shows the opposite: flat or declining life expectancy, no HIV recovery to ride (HIV was already low), and a bottom-left decliner position. Therefore, the dashboard answers CQ by letting the user verify, country by country, that the strongest gainers combined HIV reduction, investment in schooling and sustained expenditure.

5 Discussion

The dashboard achieves its design goal: each simple question can be read at a glance from a single static chart, and the complex question is answered through coordinated interaction across all four views. Three points merit critical reflection.

5.1 Strengths and Limitations

The strongest feature of the design is the cohort definition. The lagger cohort (LE < 60 in 2000) is principled and reproducible; it is computed once from the data rather than chosen by hand. Using the same cohort across SQ3 and the CQ ensures the narrative coheres. The principal limitation is a confirmation-bias risk in SQ3: by examining HIV/AIDS in isolation, the chart cannot independently demonstrate that HIV was the dominant factor rather than, say, immunisation coverage or adult mortality. A multi-factor bar chart would address this at the cost of visual clarity. The trade-off was

intentional. A secondary limitation is the dataset itself: a few indicators (alcohol, BMI) show implausible movements that appear to reflect changes in measurement units rather than real trends and were therefore excluded.

5.2 Implementation Tool Comparison

The two implementations meet the same requirements but expose different strengths. Tableau is the better tool for an analyst preparing a one-off briefing: design iteration is fast, dashboard filter actions are configured through a graphical dialog, and geographic role detection eliminates an entire engineering task. The cost is reduced control over anything Tableau does not natively support (e.g., a year-range slider that scopes time series but not the bubble chart), which must be approximated with quick filters. Plotly Dash inverts the trade-off: every layout choice and every interaction is in code, so a year-range slider, custom legends, or a programmatic country join are straightforward, but every small visual adjustment requires editing source files and restarting the server. The implementations are complementary, not competing. Tableau is the right tool for a single high-quality artefact; Dash is the right tool when the dashboard is part of a larger software system that must be version-controlled, embedded, or continuously deployed.

5.3 HCI/UCD Reflection

Three principles drove the design. First, Mackinlay's (1986) ranking of perceptual accuracy: positionon-aligned-axis was reserved for SQ1 and SQ3 (where small differences matter); colour value for SQ2; and angle/area for CQ P1, where the question concerns regional patterns and quadrant membership rather than precise magnitudes. Second, Wickens's (2008) proximity-compatibility principle: the four views are spatially adjacent because they must be mentally integrated to answer the complex question, and the filter actions preserve that integration during interaction. Third, Nielsen's (1994) heuristic of recognition over recall: every chart carries its question as its title, the cohort categories are spelt out rather than abbreviated, and the persona is named on the dashboard banner.

As a future learning objective, I plan to build a third implementation in Vega-Lite (Wilkinson 2005; Satyanarayan et al. 2017). Its grammar-of-graphics specification offers a middle ground between Tableau's UI and Dash's full Python: a JSON specification expresses each chart declaratively, and a single composition operator wires the views together.

6 Conclusion

This project produced an interactive Tableau dashboard, a coordinated Plotly Dash reimplementation, and a Tableau Story narrative for a WHO post-MDG retrospective. The analysis shows that life expectancy genuinely improved during 2000–2015 and that the gap between developed and developing nations narrowed, though the improvement was geographically uneven. For the cohort that started furthest behind, the most dramatic single change was the reduction in HIV/AIDS. The complex question is addressed through dashboard interaction: the strongest country-level gains were achieved by pairing HIV recovery with sustained schooling and health expenditure investment, while countries with none of these levers or affected by armed conflict lost ground.

In direct answer to the complex question, among the lagging countries, the strongest gains in life expectancy were achieved by those that combined all three levers: a substantial reduction in HIV/AIDS deaths, a sustained increase in years of schooling, and consistent health expenditure as a share of GDP. Countries such as Zimbabwe, Rwanda, Eritrea, Botswana and Ethiopia exemplify this combined profile and occupy the top-right gainer quadrant of CQ P1. Countries that benefited from HIV recovery alone, without parallel investments, gained less; countries affected by armed conflict during the period (Syria, Libya, Yemen) lost ground regardless of their starting position. The retrospective, therefore, points to a clear policy implication for the post-MDG era: vertical, diseasespecific interventions can deliver population-scale gains within a decade, but durable gains accrue to countries that pair disease control with broader human development investment.

References

Avert, 2024. ART scale-up and the Global Fund. [online] Available at: https://www.avert.org/professionals/hiv-around-world/global-response/funding [Accessed 10 April 2026].

Bertin, J., 1983. Semiology of Graphics: Diagrams, Networks, Maps. Madison: University of Wisconsin Press.

Few, S., 2009. Now You See It: Simple Visualization Techniques for Quantitative Analysis. Oakland: Analytics Press.

Mackinlay, J., 1986. Automating the design of graphical presentations of relational information. ACM Transactions on Graphics, 5(2), pp. 110–141.

Nielsen, J., 1994. Usability Engineering. San Diego: Morgan Kaufmann.

Rajarshi, K., 2018. Life Expectancy (WHO). [online] Kaggle. Available at: https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who [Accessed 14 April 2026].

Satyanarayan, A., Moritz, D., Wongsuphasawat, K. and Heer, J., 2017. Vega-Lite: A grammar of interactive graphics. IEEE Transactions on Visualization and Computer Graphics, 23(1), pp. 341–350.

Tufte, E.R., 2001. The Visual Display of Quantitative Information. 2nd ed. Cheshire: Graphics Press.

Wickens, C.D., 2008. Multiple resources and mental workload. Human Factors, 50(3), pp. 449–455.

Wilkinson, L., 2005. The Grammar of Graphics. 2nd ed. New York: Springer.